NITISH SATISH KADU

+16313560682 STONY BROOK, NY nitishsatish.kadu@stonybrook.edu linkedin.com/in/nitish-kadu-sbu/ nitish-portfolio-rust.vercel.app/

EDUCATION

Master of Science in Computer Science, Stony Brook University, NY, United States **Aug 2024 - May 2026**

Relevant Coursework: Distributed Systems, Analysis of Algorithm, Data Science Fundamentals, Visualization, Operating Systems

Bachelor of Engineering in Computer Engineering, University of Mumbai, Mumbai, India **Jul 2018 - May 2022**

Relevant Coursework: Microprocessors, Data Structures, Computer Networks, Software Engineering, Database Management, Distributed Computing, Machine Learning, Big Data Analytics, Natural Language Processing, Machine Learning

EXPERIENCE

Amazon Web Services, Managed Service for Apache Flink **May 2025 - Aug 2025**

Software Developer Engineer Intern *Seattle, United States*

- Led the migration of the Apache Flink patching system from a greedy distributed job scheduler to an event-driven orchestrator using AWS Step Functions and SQS, eliminating manual intervention and eradicating 200+ monthly rollbacks across 37 regions.
- Engineered a safe-deployment framework with canary testing (validated via CloudWatch logs) and an incremental blast-radius rollout, successfully patching 18,000+ applications in one AZ with zero rollbacks and reducing customer downtime by 37.5%.
- Developed idempotent, fault tolerant blue/green deployment workflows with Lambda to migrate application data planes between EC2 clusters, incorporating pre-provisioning checks to prevent resource exhaustion and ensure zero capacity-related failures.
- Designed an SLA-driven scheduling algorithm using a merge-intervals technique to dynamically compute optimal patching time window based on customers time preference, guaranteeing zero breaches of customer maintenance agreements.
- Delivered full production automation with comprehensive runbooks, transforming a manual process into a hands-off operation for the engineering team.

Jio Platforms Ltd **Jun 2022 - Jun 2024**

Software Developer *Mumbai, India*

- Led full-stack development of a scalable Service Management CRM (Angular, Spring Boot, MongoDB/PostgreSQL) that automated Incident, Service, and Change Management for 150+ organizations, handling 10,000+ monthly requests with customizable RBAC.
- Engineered AI/ML features to automate workflows and reduce ticket volume by 20,000 monthly, including an intelligent Knowledge Base for auto-suggesting solutions, a dynamic service catalog for automated request generation, and CCTV analytics for automatic incident detection at power plants.
- Boosted platform performance and reliability by cutting bundle size 30% via micro-frontends, achieving issue-free deployments with Azure DevOps CI/CD, and enhancing UX with auto-save features that reduced form abandonment.

Luna PR, Remote **May 2021 - May 2022**

Front end developer *Dubai, UAE*

- Developed and deployed 25+ web applications using React, Nextjs, Prisma, and Framer Motion for many fintech, crypto, NFT based startups by gathering requirements and delivering iterative solutions using Agile methodologies, ensuring on-time results.
- Designed a dynamic client pricing module using React, Firebase, Redux. To provide different customizable pricing options enhancing client retention, and streamlining subscription management for Luna PR and deployed on AWS EC2.
- Delivered a modern web application by utilizing Next.js features like Static Generation, Server-Side Rendering, ISR, CSR, API Routes, and Middleware, meeting business requirements and ensuring scalability, performance, and SEO optimization.

ACADEMIC PROJECTS

Crash, Fault-Tolerant Transaction System using Paxos **[GoLang, Paxos, TinyDB]**

Engineered a sharded, replicated key-value store handling intra-shard transactions via a Paxos consensus protocol and cross-shard atomic commits via a Two-Phase Commit (2PC) protocol, ensuring consistency under fail-stop faults with locking and write-ahead logging.

Byzantine Fault-Tolerant (BFT) Sharded Blockchain **[PBFT, 2PC, Python, SQLite]**

Designed a permissioned blockchain system resilient to malicious (Byzantine) actors by implementing the Practical Byzantine Fault Tolerance (PBFT) protocol for intra-shard consensus, contrasting with crash-fault tolerant systems by ensuring correctness even with adversarial nodes.

Distributed Consensus Core with Modified Paxos **[Paxos, Python, TCP/RPC]**

Architected the foundational consensus layer for a distributed bank, implementing a novel variant of the Paxos protocol where leaders aggregated uncommitted transaction logs from all servers into atomic blocks to resolve insufficient balances, ensuring strict consistency across 5 nodes under crash-fault conditions.

SKILLS

Language, Frameworks & Tools: Next JS,NodeJS, GraphQL , MongoDB, MySQL, Oracle, SQLite, PostgreSQL, MS SQL Server, H2, MariaDB, RDS , Apache Kafka, RabbitMQ, AWS SQS, Docker, Kubernetes, Jenkins, Maven, Gradle, AWS (S3, Lambda, Step Functions, Kinesis, IAM, Neptune), Azure, Git, Apigee, Flutter, Material UI, Bootstrap, Hydra,